

document_qa_modules

Project: Question answer from a uploaded document using streamlit for user interface, pinecone for generating embeddings and deepset-roberta-base-squad2 model for dealing with user question and generate answer the question.

Just to quick check on app, you can use docker if you have an account:

Run this code in your terminal<docker pull datasciencejordan/document_qa>

Or

For installing dependencies, I recommend to create/open virtual environment locally on your desktop. Then follow steps below.

1. Clone the repository:

<https://github.com/Jordan-Karangala/QA_deepset-roberta-base-squad2.git>

2. cd to the project location.

3. Installation: Use requirements.txt file to install all the dependencies for this project.

4. Once the installation is done, check the files whether chunk_embed.py, data_files_manager.py, streamlit_launcher.py are present in the folder where cloning s done.

5. Remove any files in the folders: previous_uploaded_files and uploads

Code is ready to use if you have already created an account with streamlit and pinecone.

If Not:

PINECONE Setup:

Install Pinecone

Get started with Pinecone. First, install the Python SDK:

pip install pinecone

Go to: <<https://www.pinecone.io/>> and signup.

You will have a default API key generated by pinecone which can be used later in the project or you can create a new api by clicking on create API Key.

In the project folder, open chunk_embed.py file and look for lines below to eneter your api key.

<pc = Pinecone(api_key="Enter/your/api/key/here")>. save and close the file.

check if streamlit is installed or not:

If not:

pip install streamlit

Your are all set to go now.

Testing/Running the model:

If you want to use your terminal for initiating the app, use the following command in your terminal.

```
<# streamlit run path/to/your/main.py #>
```

If you want to run from PyCharm IDE, follow the steps bellow.

Open your Streamlit app file (e.g., app.py) in PyCharm.

Edit the app to run using Python code: Replace the usual terminal-based streamlit run command with this in your code:

```
import os
```

```
if __name__ == "__main__":
```

```
    os.system("streamlit run your_app.py")
```

Configure PyCharm Run Settings:

In PyCharm, go to Run > Edit Configurations.

Click on the "+" icon to add a new configuration and choose Python.

Name it (e.g., Streamlit).

Set the Script path to your your_app.py file.

Set the Python interpreter to your desired environment (ensure Streamlit is installed there).

Click Apply and OK.

Now, you can run your Streamlit app using the PyCharm run button.